Cutoff of the simple exclusion process with inhomogeneous conductances

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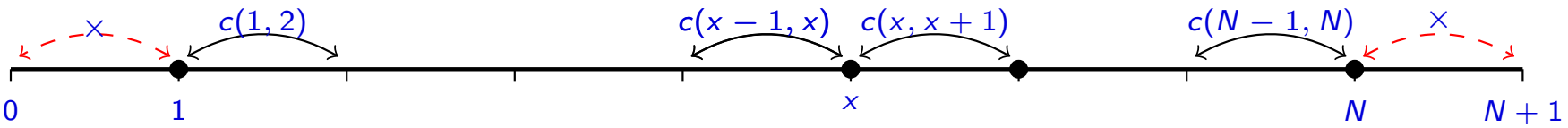
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Setup: Simple Exclusion Process with inhomogeneous conductance

Setup: Simple Exclusion Process (SEP)

Conductances: $(c(x, x+1))_{x\in\mathbb{N}}$ with values in $(0, \infty)$.
SEP with k particles in $[\![1, N]\!]$ with conductances $c(x, x+1)_{x\in\mathbb{N}}$.



(A) Each site is occupied by at most one particle (*the exclusion rule*).

(B) At each edge $\{x, x+1\}$ with $1 \le x < N$, we place a Poisson clock with rate $c(x, x+1) > 0$. When a clock rings, we swap the contents of the two sites.

Setup

- State space (1: particle 0: empty site.)

$$\Omega_{N,k} := \left\{ \xi : [\![1, N]\!] \to \{0, 1\} \;\middle|\; \sum_{i=1}^{N} \xi(i) = k \right\} .$$

- Generator: $(f : \Omega_{N,k} \mapsto \mathbb{R})$

$$\left(\mathcal{L}_{N,k} f\right)(\xi) := \sum_{i=1}^{N-1} c(i, i+1) \left[f(\xi \circ \tau_{i,i+1}) - f(\xi) \right] ,$$

  where $\tau_{i,j}$ is the transposition of the two elements i and j.

- Uniform prob measure $\mu_{N,k}$ satisfies the detailed balance condition:

$$\mu_{N,k}(\xi)\mathcal{L}_{N,k}(\xi, \eta) = \mu_{N,k}(\eta)\mathcal{L}_{N,k}(\eta, \xi),$$

  then $\mu_{N,k}$ is the invariant prob measure.

Setup

- Distance to equilibrium

$$d_{N,k}(t) := \max_{\xi \in \Omega_{N,k}} \|P_t^\xi - \mu_{N,k}\|_{\mathrm{TV}} .$$

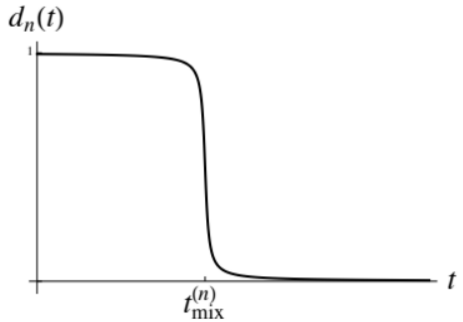
P_t^ξ: marginal distribution at instant t of the chain starting with ξ.

- ε-mixing time

$$t_{\mathrm{mix}}^{N,k}(\varepsilon) := \inf\{t \geq 0 : d_{N,k}(t) \leq \varepsilon\} .$$

- Cutoff: for all $\varepsilon \in (0,1)$,

$$\lim_{N\to\infty} \frac{t_{\mathrm{mix}}^{N,k}(\epsilon)}{t_{\mathrm{mix}}^{N,k}(1-\epsilon)} = 1 .$$



Setup

- : Spectral gap $\mathrm{gap}_{N,k}$: minimal nonzero eigenvalue of $-\mathcal{L}_{N,k}$

$$\mathrm{gap}_{N,k} := \inf_{f\,:\,\mathrm{Var}_{\mu_{N,k}}(f)>0} \frac{-\langle f, \mathcal{L}_{N,k} f\rangle_{\mu_{N,k}}}{\mathrm{Var}_{\mu_{N,k}}(f)}$$

where $\mathrm{Var}_{\mu_{N,k}}(f) := \langle f, f\rangle_{\mu_{N,k}} - \langle f, \mathbf{1}\rangle^2_{\mu_{N,k}}$.

- Relation between spectral gap and mixing time/distance to equilibrium:

$$\frac{1}{\mathrm{gap}_{N,k}} \log\frac{1}{2\varepsilon} \le t^{N,k}_{\mathrm{mix}}(\varepsilon) \le \frac{1}{\mathrm{gap}_{N,k}} \log\frac{1}{2\varepsilon\mu_{\min}}$$

where $\mu_{\min} := \min_{\xi\in\Omega_{N,k}} \mu_{N,k}(\xi)$.

$$\lim_{t\to\infty} \frac{1}{t} \log d_{N,k}(t) = -\mathrm{gap}_{N,k}\,.$$

Question: How does the disordered setup (inhomogeneous conductances) affect the system in terms of spectral gap/mixing time?

Previous Results

Homogeneous conductances $c(x, x+1) \equiv 1$ for one particle $(k = 1)$

- Spectral gap
$$\mathrm{gap}_{N,1} = 2(1 - \cos(\pi/N)) = (1 + o(1))\pi^2/N^2 .$$
- Eigenfunctions
$$g_i^{(N)}(x) := \cos(i\pi(x - 1/2)/N), \quad 0 \le i < N .$$
- Eigenvalues
$$-\lambda_i^{(N)} = -2\,(1 - \cos(i\pi/N)) \quad \mathcal{L}_{N,1} g_i^{(N)} = -\lambda_i^{(N)} g_i^{(N)} .$$

Homogeneous conductances $c(x, x+1) \equiv 1$ for many particles

- [Aldous] [Wilson] [Lacoin] Assuming $\liminf_{N\to\infty} \min(k, N-k) = \infty$,
$$t_{\mathrm{mix}}^{N,k}(\varepsilon) = (1 + o(1))\frac{N^2}{2\pi^2}\log k , \quad \mathrm{gap}_{N,k} = \mathrm{gap}_{N,1} = (1 + o(1))\frac{\pi^2}{N^2} .$$

Previous results

Inhomogeneous conductance $c(x, x+1) > 0$

- Aldous' spectral gap conjecture (Proved by [Caputo, Liggett, Richthammer, JAMS '10]) implies:

$$\mathrm{gap}_{N,k} \;=\; \mathrm{gap}_{N,1}\,, \quad \forall\, k \in [\![1, N-1]\!]\,.$$

- A function $f : [\![1, N]\!] \mapsto \mathbb{R}$ is j-monotone if it displays exactly $(j-1)$ distinct local extrema in $[\![2, N-1]\!]$.
  Nodal domains:

$$\#\text{connected components of } \{x \in [\![1, N]\!], f(x) \neq 0\}\,.$$

- [Miclo]: $L_{N,1} g_i^{(N)} = -\lambda_i^{(N)} g_i^{(N)}$ with $0 = \lambda_0^{(N)} < \lambda_1^{(N)} < \cdots \lambda_{N-1}^{(N)}$
  $g_i^{(N)}$ is i-monotone and has $i+1$ nodal domains.

Our results

Proposition (Y. '25)

For any positive conductances $(c(x,x+1))_{x\in\mathbb{N}}$, $g_1^{(N)}$ is strictly monotone.

Write $r(x,x+1) := 1/c(x,x+1)$ and $r(n,m) := \sum_{x=n}^{m-1} r(x,x+1)$.
Assume (LLN) condition

$$\limsup_{N\to\infty} \frac{1}{N} \sup_{2\le m\le N} \left|(r^{(N)}(1,m) - (m-1)\right| \;=\; 0\,. \qquad \textbf{(LLN)}$$

When $(r^{(N)}(x-1,x))_{2\le x\le N}$ is IID with expectation $\mathbb{E}[r(x,x+1)] = 1$, by the strong LLN we have

$$\mathbb{P}\left(\lim_{N\to\infty} \frac{1}{N} \max_{2\le m\le N} |r(1,m) - (m-1)| = 0\right) \;=\; 1\,.$$

Our results

Theorem (Faggionato '08, Y. '25)

If the (LLN) condition on the resistances holds, we have

$$\lim_{N\to\infty} \frac{N^2 \mathrm{gap}_N}{\pi^2} = 1\,.$$

Furthermore, concerning the shape and (weighted) derivative of the eigenfunction g_1 with $g_1(1) := 1$ corresponding to the spectral gap, i.e. $\mathcal{L}_{N,1} g_1 = -\mathrm{gap}_N \cdot g_1$ and setting

$$h(x) := \cos\left(\frac{\pi(x-1/2)}{N}\right), \quad \forall\, x \in [\![1, N]\!]\,,$$

we have ($(c\nabla f)(x) := c(x-1,x)[f(x) - f(x-1)]$)

$$\lim_{N\to\infty} \sup_{x\in[\![1,N]\!]} |g_1(x) - h(x)| = 0\,,$$

$$\lim_{N\to\infty} \sup_{x\in[\![1,N]\!]} |N(c\nabla g_1)(x) - N(\nabla h)(x)| = 0\,.$$

Our results

Remark

The method in the forementioned theorem also works for the other j-monotone eigenfunctions under the (LLN) assumption, i.e. with $K_0 \in \mathbb{N}$ being any prefixed constant, for all $1 \leq i \leq K_0$,

$$\lim_{N\to\infty} |\lambda_i N^2 - \pi^2 i^2| \;=\; 0\,,$$

$$\lim_{N\to\infty} \sup_{x\in[\![1,N]\!]} \left| g_i(x) - \cos\left(\frac{i\pi(x-1/2)}{N}\right)\right| \;=\; 0\,,$$

$$\lim_{N\to\infty} \sup_{x\in[\![1,N]\!]} |N(c\nabla g_i)(x) - N(\nabla h_i)(x)| \;=\; 0\,,$$

where $g_i(1) = 1$.

Our results: mixing time

Assumption

Exist constants $\upsilon \in (0,1)$ *and* $C_{\mathbb{P}} > 0$, *a sequence of positive numbers* $\left(\bar{\Upsilon}_N\right)_N > 0$ *with* $\lim_{N\to\infty} \bar{\Upsilon}_N = 0$ *and* $\lim_{N\to\infty} \bar{\Upsilon}_N \log N = \infty$ *such that*

$$\max_{1\le x<N} r(x,x+1) \;\le\; C_{\mathbb{P}} \exp\left((\log N)^{\upsilon}\right),$$

$$\min_{1\le x<N} r(x,x+1) \;\ge\; \bar{\Upsilon}_N .$$

Exists $\varrho \in (0,1]$ *and* $c_\varrho > 0$ *such that*

$$c_\varrho N^{\varrho} \;\le\; k_N \;\le\; N/2 .$$

Theorem (Y. '26+ AIHP)

Under (LLN) and the assumption above, for all $\varepsilon \in (0,1)$ *we have*

$$\lim_{N\to\infty} \frac{2\pi^2 t_{\mathrm{mix}}^{N,k}(\varepsilon)}{N^2 \log k_N} \;=\; 1 .$$

Outline

- Idea for the spectral gap
- Idea for the shape & derivative of eigenfunction
- Idea for the lower bound on the mixing time
- Idea for the upper bound on the mixing time

Idea: the spectral gap

Idea: the spectral gap

Setting $c(N, N+1) = 1$, for $\lambda > 0$, define $f^\lambda : [\![0, N+1]\!] \mapsto \mathbb{R}$ by $f^\lambda(0) = f^\lambda(1) = 1$ and for $x \in [\![1, N]\!]$,

$$f^\lambda(x+1) \;=\; f^\lambda(x) + \frac{1}{c(x, x+1)} \Big[(c\nabla f^\lambda)(x) - \lambda f^\lambda(x) \Big] \,.$$

Note that (the restriction to $[\![1, N]\!]$ of) f^λ is an eigenfunction of $\mathcal{L}_{N,1}$ if and only if

$$f^\lambda(N+1) \;=\; f^\lambda(N) \,.$$

There is no eigenfunction satisfying $f^\lambda(1) = 0$ or $f^\lambda(N) = 0$.

Idea: the spectral gap

For $\lambda > 0$ and $x \in [\![1, N+1]\!]$, we set

$$b(\lambda, x) \;:=\; -\frac{(c\nabla f^{\lambda})(x)}{f^{\lambda}(x-1)}$$

convention: $b(\lambda, x) = \overline{\infty}$ if $f^{\lambda}(x-1) = 0$, and $\overline{\mathbb{R}} = \mathbb{R} \cup \{\overline{\infty}\}$. We have

$$b(\lambda, x+1) \;=\; \frac{b(\lambda, x)}{1 - c(x-1, x)^{-1} b(\lambda, x)} + \lambda\,.$$

Idea for the spectral gap

Setting $B^{(N)}(x) := b(\lambda, x)N$ and $\lambda := \alpha/N^2$, we have

$$B^{(N)}(x+1) = \frac{B^{(N)}(x)}{1 - N^{-1} r^{(N)}(x-1, x) B^{(N)}(x)} + \frac{\alpha}{N},$$

which starts from $B^{(N)}(1) := 0$.

$$N\left[B(x+1) - B(x)\right] = \frac{r(x-1, x) B(x)^2}{1 - B(x)\, r(x-1, x) N^{-1}} + \alpha$$

Intuition: the asymptotic ODE

$$\begin{cases} \frac{\mathrm{d}y}{\mathrm{d}x} = y^2 + \alpha, & x \in (0, 1) \\ y(0) = 0. \end{cases}$$

Its unique solution:

$$y(x) = \sqrt{\alpha} \tan\left(\sqrt{\alpha} \cdot x\right).$$

Therefore $\alpha = i^2 \pi^2$.

Idea: shape/derivative of the eigenfunction

Idea for shape of the eigenfunction

Eigenfunction corresponding to the spectral gap when $r(j-1,j) \equiv 1$

$$h(x) \;=\; h_N(x) \;=\; \cos\left(\frac{\pi(x-1/2)}{N}\right).$$

The spectral gap is

$$\overline{\lambda} \;:=\; 2\left[1-\cos\left(\frac{\pi}{N}\right)\right] \;=\; \frac{\pi^2}{N^2}+O\left(\frac{1}{N^4}\right).$$

By $b(\lambda,x) \;=\; -\frac{(c\nabla g)(x)}{g(x-1)}$ and $b(\lambda,x)=B(x)/N$, for $x \geq 2$ we have

$$g(x) \;=\; \left[1-r(x-1,x)N^{-1}B\left(x\right)\right]g(x-1).$$

Writing $u(x) := h(x)-g(x)$, we have

$$\begin{aligned} u(x) \;=\; u(x-1)&\left[1-\frac{r(x-1,x)B\left(x\right)}{N}\right] \\ &+\frac{h(x-1)}{N}\left[r(x-1,x)B\left(x\right)-\overline{B}\left(x\right)\right]. \end{aligned}$$

Iterate the equation above to conclude the proof for $x \leq N/3$.

Idea for the derivative of the eigenfunction

For $x \le N/3$: by

$$(c\nabla g_i)(x+1) \;=\; -\lambda_i \sum_{y=1}^{x} g_i(y)\,,$$

we have

$$\begin{aligned}
&|N(c\nabla g)(x) - N(\nabla h)(x)| \\
&= \left| -N\lambda_1 \sum_{k=1}^{x-1} g(k) + N\overline{\lambda} \sum_{k=1}^{x-1} h(k) \right| \\
&\le \left| -N\lambda_1 \sum_{k=1}^{x-1} [g(k) - h(k)] \right| + \left| N\left(\overline{\lambda} - \lambda_1\right) \sum_{k=1}^{x-1} h(k) \right| \\
&\le N\lambda_1 \sum_{k=1}^{x-1} |[g(k) - h(k)]| + N|\overline{\lambda} - \lambda_1|(x-1)\,.
\end{aligned}$$

For $N/3 \le x \le 2N/3$: use $A(x) = 1/B(x)$

$$A(x)\left[1 - \frac{g(x)}{g(x-1)}\right] \;=\; r(x-1,x)N^{-1}\,.$$

Idea: the lower bound on the mixing time

Idea for the lower bound on the mixing time

$$\mathrm{F}(\xi) = \mathrm{F}_1(\xi) = \sum_{1 \le x \le N} \xi(x) g_1(x)$$

is an eigenfunction satisfying $\mathcal{L}_{N,k}\mathrm{F} = -\mathrm{gap}_N \cdot \mathrm{F}$. For $t_0 > 0$, define

$$F(t,\xi) := e^{\lambda_1 (t-t_0)} \mathrm{F}(\xi)\,, \quad \forall\, \xi \in \Omega_{N,k}\,,$$

and study the Dynkin martingale

$$M_t \;:=\; F(t,\eta_t^\nu) - F(0,\eta_0^\nu) - \int_0^t (\partial_s + \mathcal{L}_{N,k})\, F(s,\eta_s^\nu)\, \mathrm{d}s\,.$$

$$\mathbf{E}\left[\mathrm{F}\left(\eta_{t_0}^\nu\right)\right] \;=\; \mathbf{E}\left[F\left(t_0,\eta_{t_0}^\nu\right)\right] \;=\; \mathbf{E}\left[F(0,\eta_0^\nu)\right] \;=\; e^{-\lambda_1 t_0}\mathbf{E}\left[\mathrm{F}\left(\eta_0^\nu\right)\right]\,.$$

$$\mathbf{E}[M_{t_0}^2] = \mathbf{E}[\int_0^{t_0} \partial_s \langle M\rangle_s \mathrm{d}s]$$

$$\overline{\eta}_t^\nu(x,x+1) := \eta_t^\nu(x)\,(1-\eta_t^\nu(x+1)) + \eta_t^\nu(x+1)\,(1-\eta_t^\nu(x))$$

$$\partial_t \langle M_\cdot\rangle_t \;=\; e^{2\lambda_1(t-t_0)} \sum_{x=1}^{N-1} \overline{\eta}_t^\nu(x,x+1) r(x,x+1) \left[c(x,x+1)(g(x)-g(x+1))\right]^2\,.$$

Idea for the lower bound on the mixing time

At equilibrium

$$\mathbf{E}\left[\mathrm{F}(\eta_{t_0}^{\mu})\right] = \mu_{N,k}(\mathrm{F}) = \frac{k}{N}\sum_{1\le x\le N} g_1(x) = 0\,, \qquad \mathrm{Var}_{\mu}(\mathrm{F}) \asymp k\,.$$

1. If ν concentrates at one configuration, then
$$\mathbf{E}\left[\mathrm{F}(\eta_{t_0}^{\nu}) - \mathbf{E}\left[\mathrm{F}(\eta_{t_0}^{\nu})\right]\right]^2 = \mathbf{E}\left[F(t_0,\eta_{t_0}^{\nu}) - F(0,\eta_0^{\nu})\right]^2 = \mathbf{E}\left[M_{t_0}^2\right]\,.$$
2. If ν is non-degenerated, we have
$$\begin{aligned}&\mathbf{E}\left[\mathrm{F}(\eta_{t_0}^{\nu}) - \mathbf{E}\left[\mathrm{F}(\eta_{t_0}^{\nu})\right]\right]^2\\ =\ &\mathbf{E}\left[F(t_0,\eta_{t_0}^{\nu}) - F(0,\eta_0^{\nu}) + F(0,\eta_0^{\nu}) - \mathbf{E}\left[F(0,\eta_0^{\nu})\right]\right]^2\\ \le\ &2\mathbf{E}\left[M_{t_0}^2\right] + 2\mathbf{E}\left[F(0,\eta_0^{\nu}) - \mathbf{E}\left[F(0,\eta_0^{\nu})\right]\right]^2\,.\end{aligned}$$

If $N/64 \le k \le N/2$, take $\nu = \delta_{\wedge}$.

If $(\log N)^{1+\gamma} \le k < N/64$, take ν as follows: sample a configuration according to $\mu_{N,2k}$, keep the first k particles and project the rest to be empty sites.

An explanation for the lower bound $t_{\mathrm{mix}} \ge \frac{1}{2\lambda_1}\log k - \frac{C}{\lambda_1}$:

$$k e^{-\lambda_1\cdot\left(\frac{1}{2\lambda_1}\log k - \frac{C}{\lambda_1}\right)} = e^{C}\sqrt{k}.$$

Idea: the upper bound on the mixing time

Idea: the upper bound on the mixing time

Height function:

$$\xi \in \Omega_{N,k} \quad \to \quad h^{\xi}(x) \;:=\; \sum_{y=1}^{x} \xi(y) - \frac{k}{N} x \,.$$

A partial order:

$$\left(\xi \;\leq\; \xi'\right) \quad \Leftrightarrow \quad \left(h^{\xi}(x) \;\leq\; h^{\xi'}(x)\,, \forall\, x \in [\![1, N]\!]\right) \,.$$

Attractive:

$$\left(h^{\xi} \leq h^{\xi'}\right) \quad \Rightarrow \quad \left(\forall\, t \geq 0,\; h_t^{\xi} \leq h_t^{\xi'}\right) \,.$$

Coalescing time:

$$\begin{aligned} T_1 \;&:=\; \inf\left\{t \geq 0 :\; h_t^{\wedge} = h_t^{\mu}\right\} \,, \\ T_2 \;&:=\; \inf\left\{t \geq 0 :\; h_t^{\vee} = h_t^{\mu}\right\} \,. \end{aligned}$$

Idea: the upper bound on the mixing time

Construct a supermartingale: inspired by [Wilson, '04], embed the segment $[\![1, N]\!]$ in $[\![-\lfloor \delta N \rfloor, N + \lfloor \delta N \rfloor]\!]$ and place conductance $(c(x, x+1) = 1)_{x \notin [1, N-1]}$. The principle eigenfunction satisfies:

$$\lim_{N \to \infty} \sup_{x \in [\![-\lfloor \delta N \rfloor, N + \lfloor \delta N \rfloor]\!]} \left| G(x) - \cos\left(\frac{\pi(x + \lfloor \delta N \rfloor + 1/2)}{\bar{N}} \right) \right| = 0 .$$

Define $\bar{G}(x) := G(x) - G(x+1) > 0$ and

$$\mathbf{F}(\xi) := \sum_{x=1}^{N-1} h^{\xi}(x) \bar{G}(x) .$$

For $\xi, \xi' \in \Omega_{N,k}$ with $\xi \leq \xi'$, since $h^{\xi}(x) \leq h^{\xi'}(x)$ and $\bar{G}(x) > 0$,

$$\mathbf{F}(\xi) \leq \mathbf{F}(\xi') .$$

Furthermore, if $\xi \leq \xi'$ with $\xi \neq \xi'$, we have $\mathbf{F}(\xi) < \mathbf{F}(\xi')$.

Idea: upper bound on the mixing time

Using $h^\xi(0) = h^\xi(N) = 0$ and for $x \in [\![1, N-1]\!]$

$$\begin{aligned}\left(\mathcal{L}_{N,k} h^\xi\right)(x) &= c(x, x+1)\left[\xi(x+1) - \xi(x)\right] \\ &= c(x, x+1)\left[\left(h^\xi(x+1) - h^\xi(x)\right) - \left(h^\xi(x) - h^\xi(x-1)\right)\right]\end{aligned}$$

we obtain

$$\begin{aligned}\left(\mathcal{L}_{N,k}\mathbf{F}\right)(\xi) &= \sum_{x=1}^{N-1} \bar{G}(x)\left(\mathcal{L}_{N,k} h^\xi\right)(x) \\ &= -\bar{\lambda}_1 \mathbf{F}(\xi) - h^\xi(1) c(0,1)\bar{G}(0) - h^\xi(N-1) c(N, N+1)\bar{G}(N)\end{aligned}$$

where $\bar{\lambda}_1$ is the spectral gap of the system in the longer line segment.

Idea: upper bound on the mixing time

For $\xi \leq \xi'$,

$$\begin{aligned}
&\left(\mathcal{L}_{N,k}\mathbf{F}\right)(\xi') - \left(\mathcal{L}_{N,k}\mathbf{F}\right)(\xi) \;=\; -\bar{\lambda}_1\left[\mathbf{F}(\xi') - \mathbf{F}(\xi)\right] \\
&- \left[h^{\xi'}(1) - h^{\xi}(1)\right] c(0,1)\bar{G}(0) - \left[h^{\xi'}(N-1) - h^{\xi}(N-1)\right] c(N,N+1)\bar{G}(N) \\
&\leq\; -\bar{\lambda}_1\left[\mathbf{F}(\xi') - \mathbf{F}(\xi)\right] .
\end{aligned}$$

Then $(\mathbf{F}(h_t^{\wedge}) - \mathbf{F}(h_t^{\mu}))_{t\geq 0}$ is a supermartingale with decay rate $\bar{\lambda}_1$.

Combine the approachs in [Lacoin AOP'16] and [Labbé, Lacoin AAP'20] to adapt to the disordered setup to conclude the proof.


Diolch
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